Did Meridian's outreach work?

It raised enrollment by about 11.5 points.

Its effect on surgery can't be detected from the historical data: that needs the pilot.

The bottom line

+11.5

points of enrollment within 180 days, 89% interval [+10.6, +12.4]

+0.05

points of surgery within 12 months, best-controlled estimate [−0.42, +0.61]

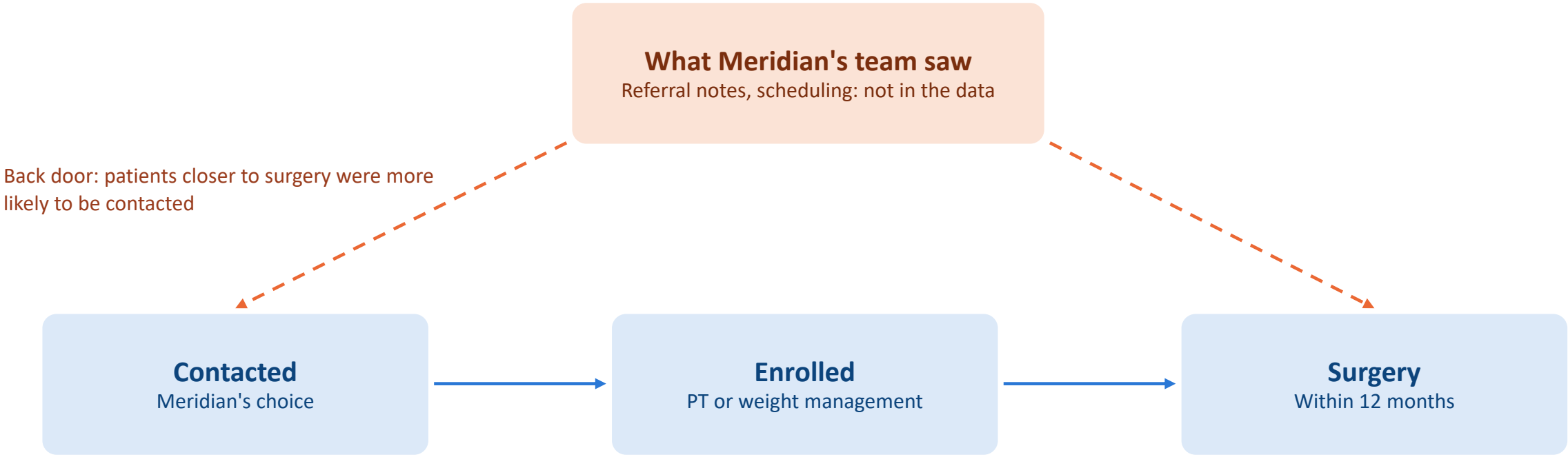
45%

posterior probability that outreach reduces surgery: a coin flip

What this means for Meridian

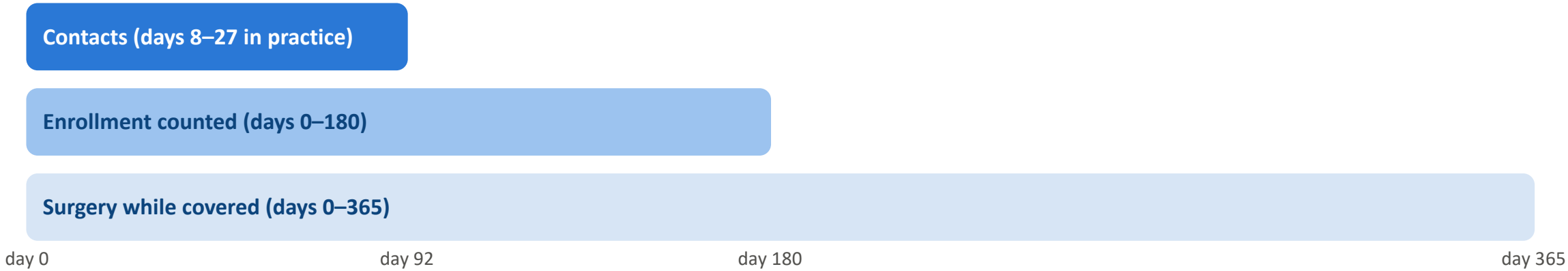
- Outreach works at the first step: contacted patients enrolled 25% → 37% of the time.
- Whether that enrollment prevents surgery can't be read from history. Meridian chose who to contact using information that isn't in the data, and the plausible effect is smaller than the remaining uncertainty.
- The randomized holdout in the pilot, followed for 12 months, is the only way to measure the surgery effect.

Why the simple comparison misleads



Raw comparison: contacted patients had surgery 3.5% of the time vs 2.7% for everyone else. That isn't outreach causing surgery. It's the dashed path: Meridian contacted sicker patients.

Emulate the trial Meridian didn't run, once per wave



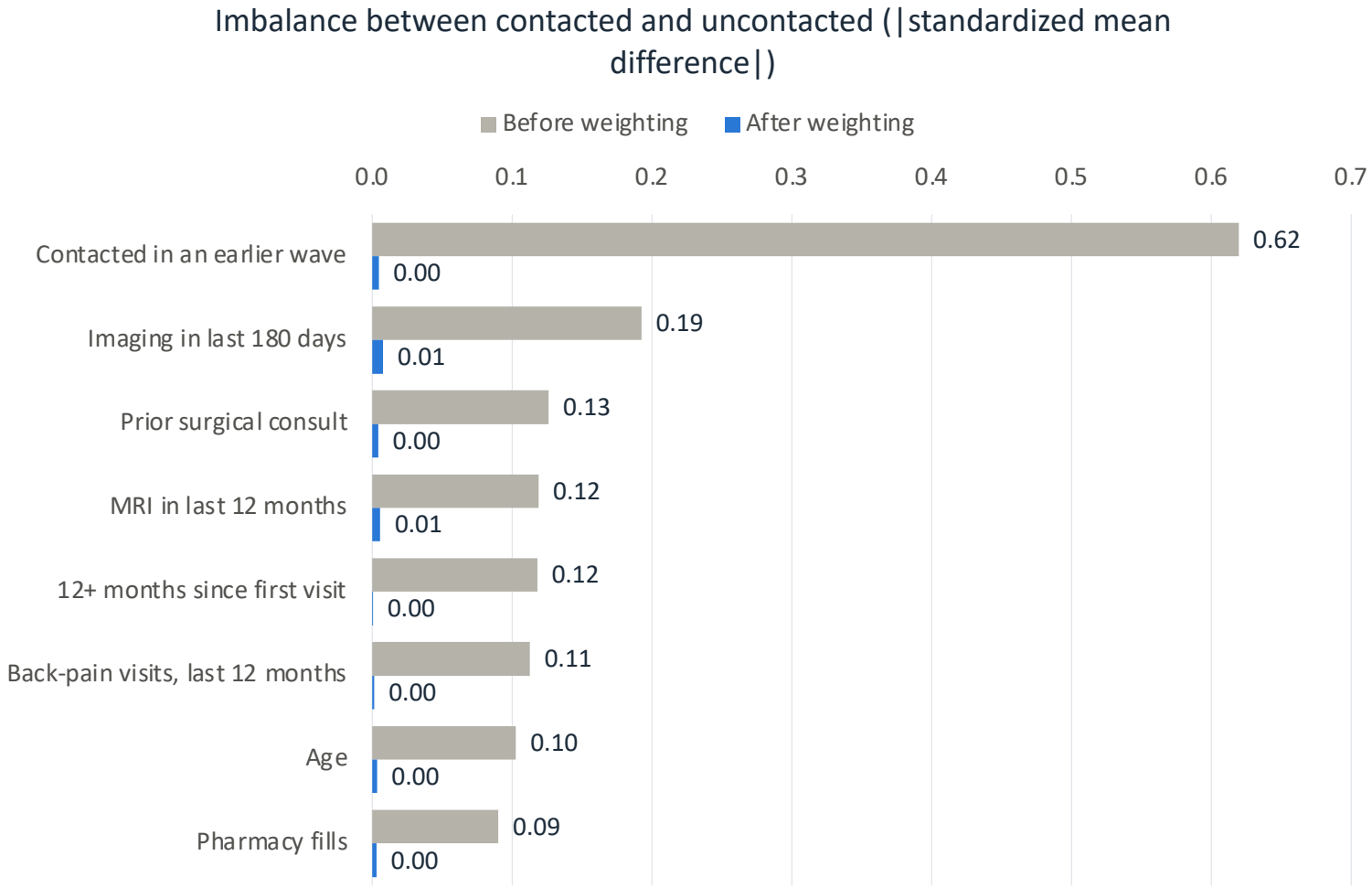
Time zero = wave start for both groups, so nobody has to stay surgery-free to count as uncontacted.

Wave	Eligible	Contacted	Share contacted
2024-07	10,787	2,490	23%
2025-01	16,477	2,753	17%
2025-07	19,182	2,399	13%

Eligible at wave start

- Covered, back-pain history, no surgery yet
- No surgical consult in the last 90 days
- Never opted out of outreach
- Adjusted for 37 covariates: the risk-model features plus earlier outreach

Weighting makes the groups comparable on what we see



Who Meridian contacted
A model of who got contacted reaches AUC 0.69. Meridian mostly avoided re-contacting earlier-wave patients, and favored patients with recent imaging, consults and MRIs.

After weighting
Every covariate is within 0.01 (0.10 is the usual flag), and the comparison group keeps an effective 26,389 patients.

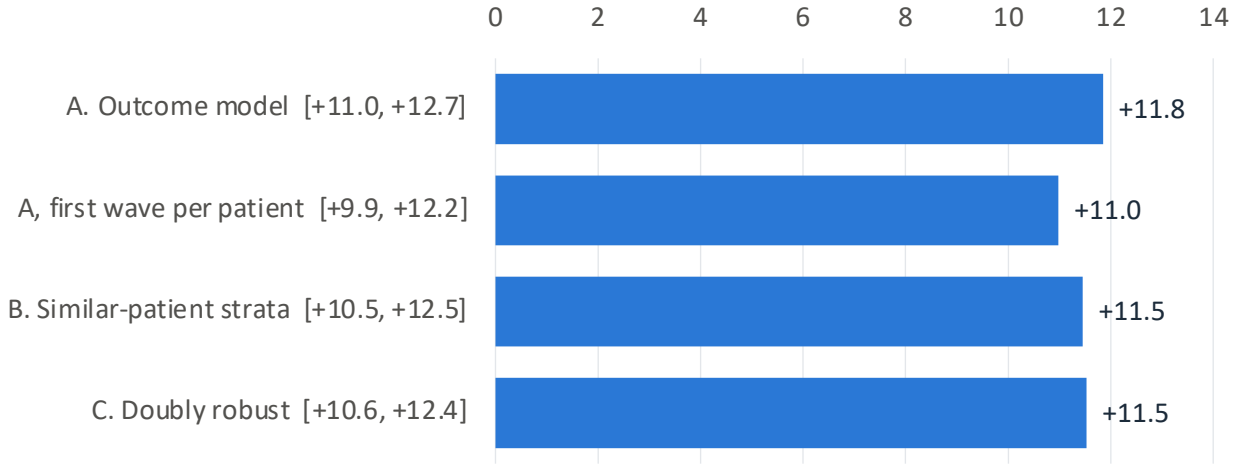
The catch
Balance on what we see says nothing about what Meridian's team saw.

Enrollment

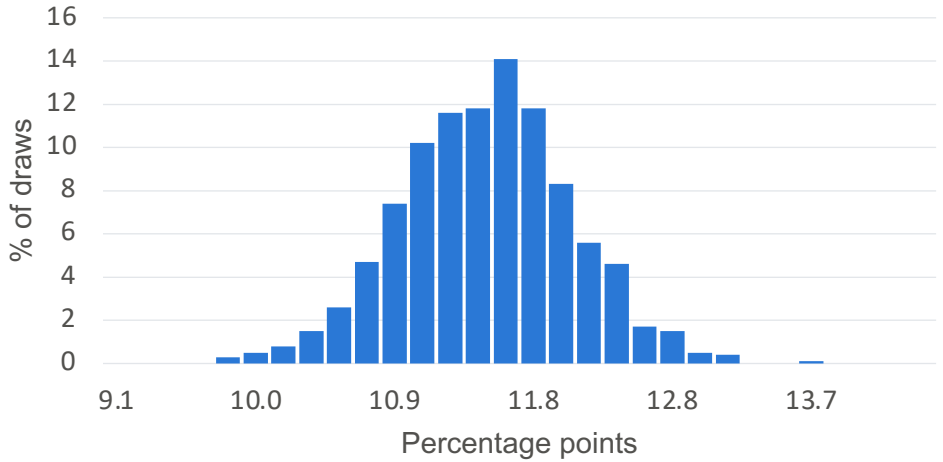
The first link in the chain: does contact get patients into a program?

Outreach raises enrollment by about 12 points

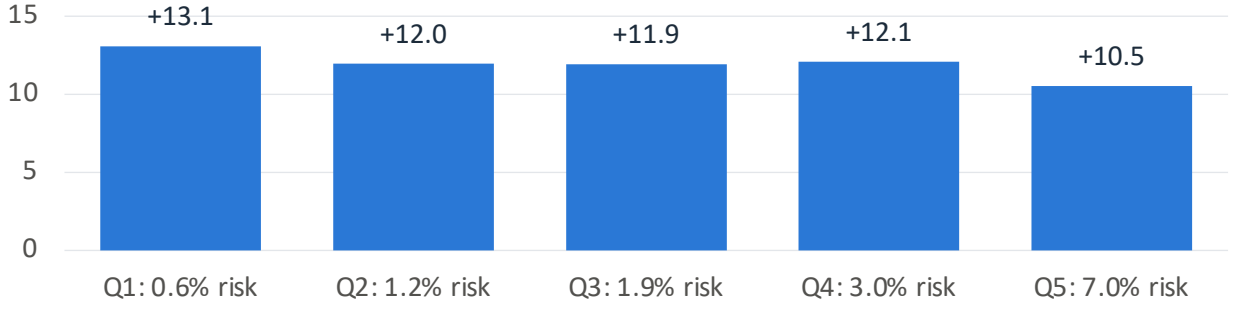
Effect on 180-day enrollment, percentage points (89% interval)



Posterior distribution of the effect (doubly robust)



By baseline surgery-risk quintile (hierarchical model)



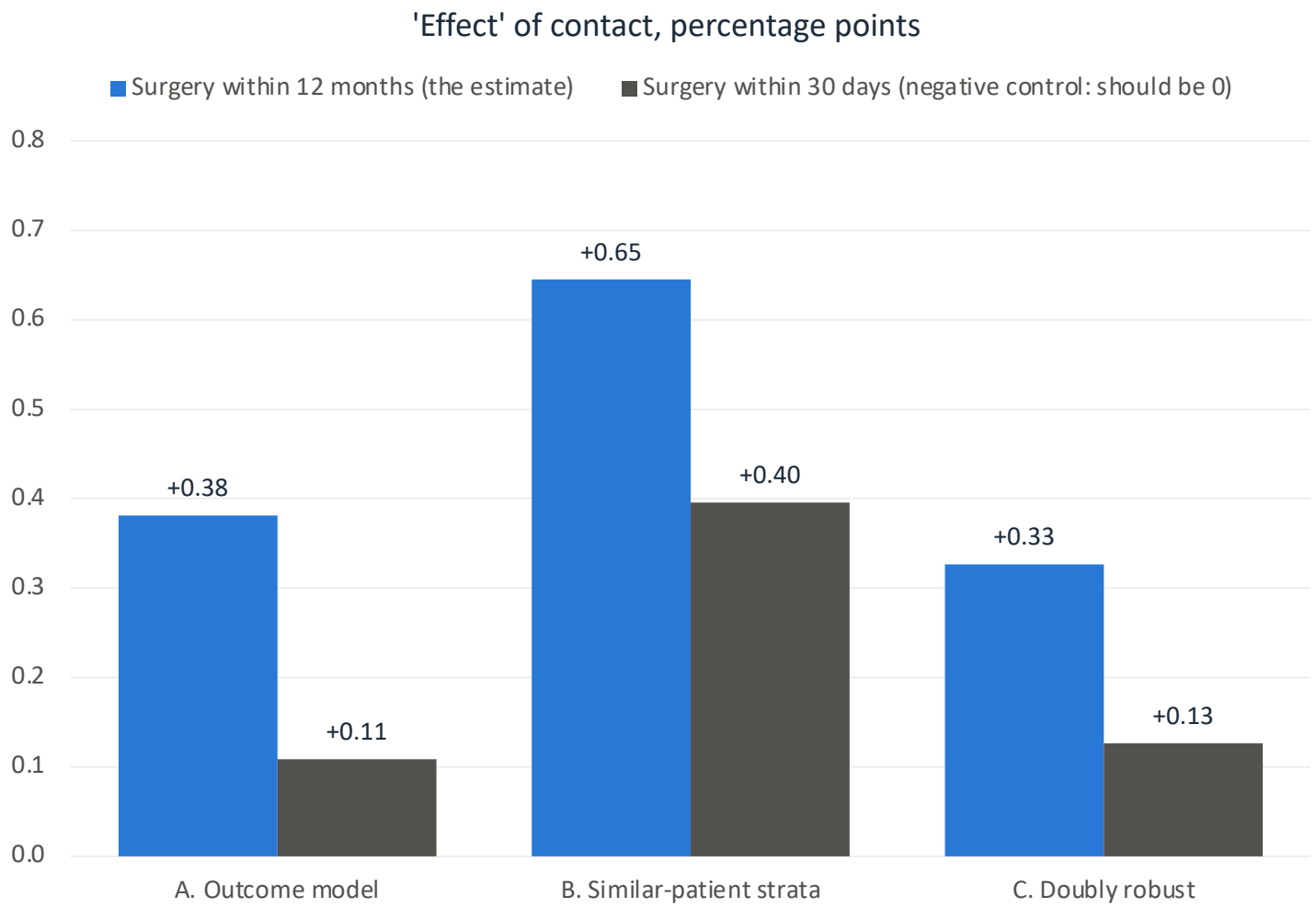
Consistent everywhere

Every approach lands at +11 to +12 points, from a 25% baseline. The lift is about the same at every risk level, slightly lower at the top. The risk score decides who's worth contacting; outreach works about equally on everyone.

Surgery

The link that matters financially: does outreach prevent surgery?

Every naive estimate comes with a matching bias



Taken at face value
outreach raises surgery by 0.33 points (doubly robust, [−0.02, +0.69]).

But look at the gray bars
Contacted patients also had more surgeries in the first 30 days, before any program could act. Outreach can't cause those. They were already in the pipeline.

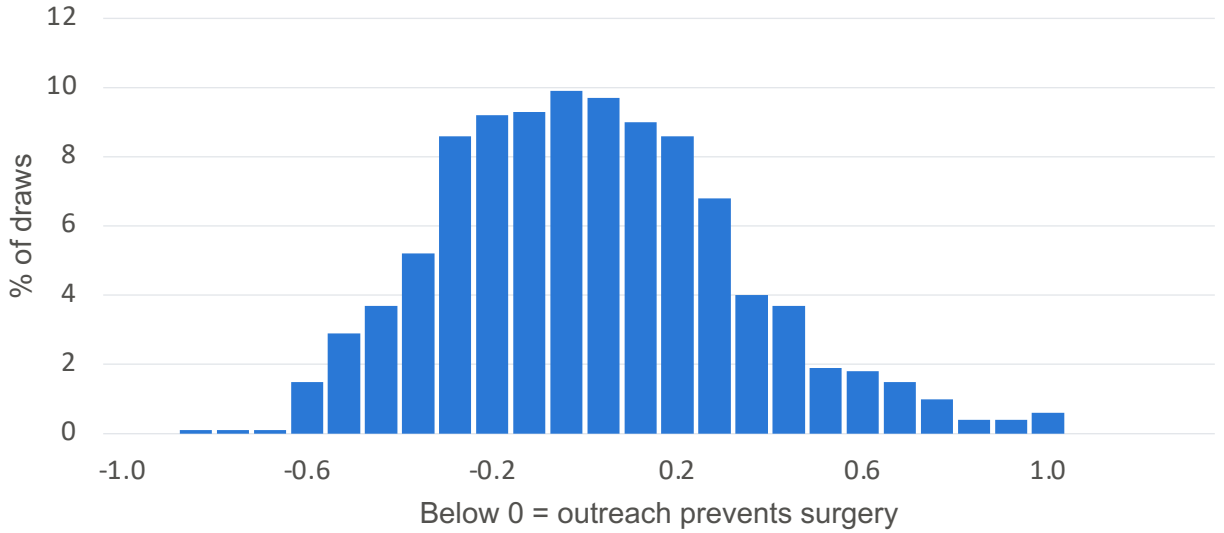
Approach B is the most biased
Ten strata per wave leave imbalance inside each stratum. The doubly robust estimate agrees with A, not B.

Restart the clock at day 30 and correct for what's left

Landmark design

- Drop the 179 surgeries already in the pipeline (days 0–30)
- Days 30–90 as a control window: too soon for a program to change a surgical decision. Risk ratio 1.10
- Days 90–365 as the main window. Risk ratio 1.11
- Corrected = main ÷ control: 1.03 [0.78, 1.33]

Posterior: corrected effect on surgery, days 90–365 (percentage points)



+0.05

points, 89% interval [−0.42, +0.61]

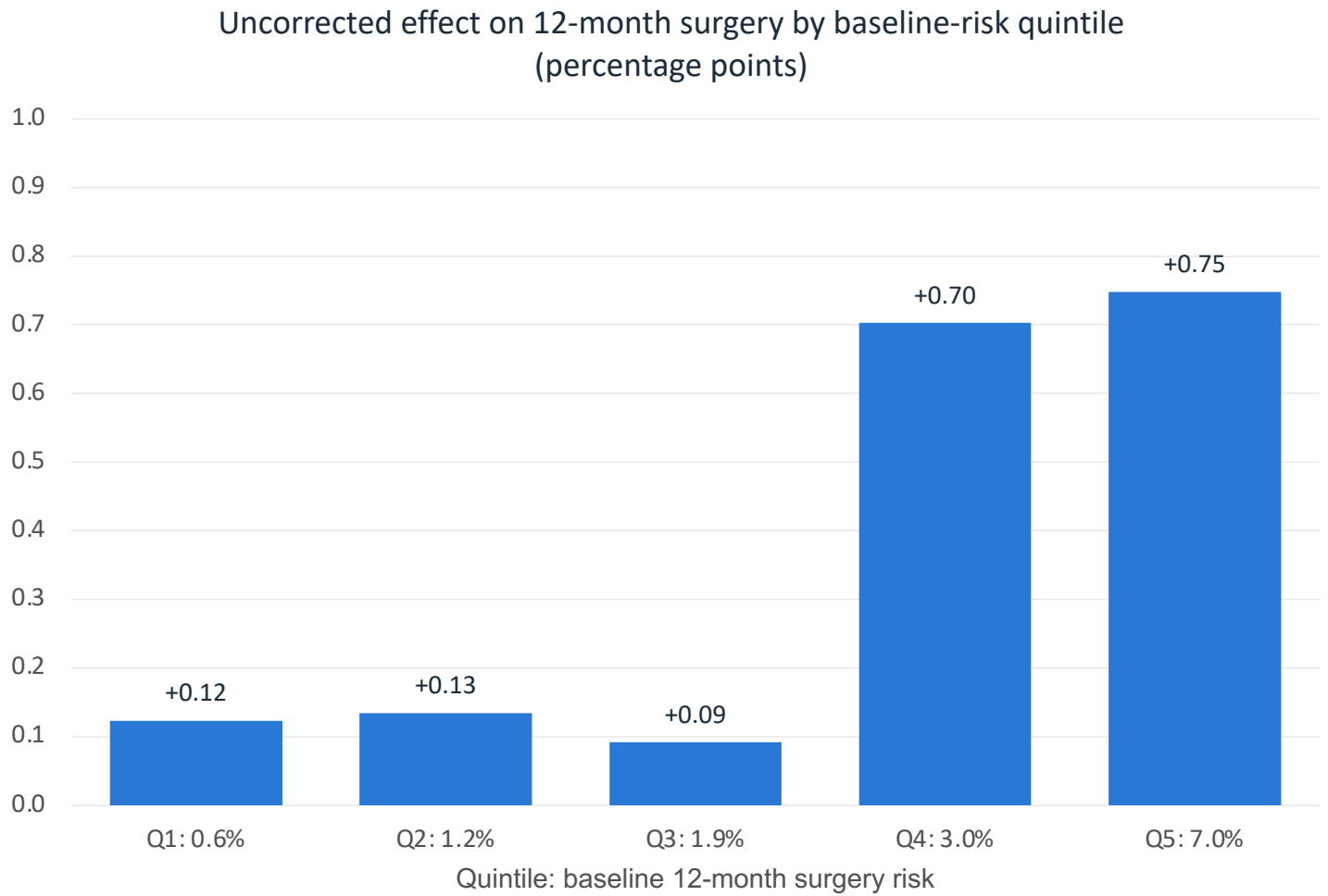
45%

posterior probability outreach is protective

1.45

E-value: a modest unmeasured confounder could explain the uncorrected estimate

Where the apparent 'effect' sits: the highest-risk patients



Concentrated in Q4–Q5
Q4 +0.70 [+0.08, +1.47], Q5 +0.75 [−0.27, +1.79]. Q1–Q3 are close to zero.

Why this points to selection
That's where clinical information outside the data, like a booked procedure, would matter most. It isn't evidence that outreach causes surgery in high-risk patients.

Three other explanations, checked and ruled out

Contacts triggered by new visits

If Meridian called patients after a new encounter mid-wave, adjusting at wave start would miss it.

Ruled out: contacted and uncontacted patients had near-identical activity before the contact date (median 54 vs 56 days since the last visit).

A hidden confounder in the extract

Cancelled consults, recent spine activity, denied claims, recent orthopedic visits.

Ruled out: all differ by under 0.06 SMD between groups. The selection is on something outside the data.

PCP-level contact rates as an instrument

Some PCPs' patients were contacted more often (F ≈ 172).

Rejected: it tracks case mix (correlates −0.24 with tenure) and would give an interval of about ±40 points.

History can't resolve an effect this small

0.1–0.3

points: the plausible surgery reduction (≈10-point enrollment lift × ~30% risk reduction × ~3% risk)

±0.5

points: half-width of the best-controlled 89% interval

+2.8

points per enrollment (Wald ratio [−0.2, +6.1]): too wide, and still biased

What it would take

- The best-controlled interval is several times wider than any realistic effect, and it still leans on an untestable assumption about the remaining selection.
- A randomized holdout removes the selection entirely. Enrollment is measurable in one quarter; surgery needs the holdout kept for 12 months and pooled across quarters.
- Use the 11.5-point enrollment lift in the value-of-the-list calculation now, and treat the surgery reduction as an assumption until the pilot reads out.

Bank the enrollment lift; let the pilot answer surgery

- Outreach raises enrollment by about 11.5 points, consistently and at every risk level.
- Its effect on surgery is undetectable from history: Meridian's targeting used information outside the data.
- Run the 80/20 randomized holdout, keep it for 12 months, and measure surgery directly.